

Problem 8.18. Let B be the language of properly nested parentheses and brackets. For example, $((()())())$ is in B but $([])$ is not. Show that B is in L .

Proof. To show that B is in L , we give a log space algorithm M that decides B .

$M =$ “On input $\langle w \rangle$, where w is a string:

1. Let p and b be binary integers.
2. Starting at start of the tape, scan for the first unmarked symbol w_i in $w = w_1w_2 \dots w_n$. If no unmarked symbol is found, then *accept*.
3. If w_i is $)$ or $]$, then *reject*.
4. Let $p = 0$ and $b = 0$.
5. Repeat for each unmarked symbol $w_j = w_{i+1}w_{i+2} \dots w_n$:
 6. If w_j is $($, then increment p by 1.
 7. If w_j is $[$, then increment b by 1.
 8. If w_j is $)$, then:
 9. If w_i is $($ and $p = 0$, then:
 10. If $b = 0$, then mark both w_i and w_j , and go to step 2. Otherwise, *reject*.
 11. If $p > 0$, then decrement p by 1. Otherwise, *reject*.
 12. If w_j is $]$, then:
 13. If w_i is $[$ and $b = 0$, then:
 14. If $p = 0$, then mark both w_i and w_j , and go to step 2. Otherwise, *reject*.
 15. If $b > 0$, then decrement b by 1. Otherwise, *reject*.
 16. *Reject*.”

In steps 2 to 4, the algorithm repeatedly picks an unmatched opening parentheses or bracket. Then, in steps 5 to 17, it tries to find the matching closing parentheses or bracket. If a match is found, then the algorithm starts all over again. The algorithm accepts if all opening parentheses and brackets are successfully matched with their corresponding closing parentheses or bracket. $\square$